

## Advanced (Habanero)

**Q1.** Solve the inequality  $2x + 5 > 11$ .

- (A)  $x > 3$
- (B)  $x < 3$
- (C)  $x > -3$
- (D)  $x < -3$
- (E)  $x = 3$

**Q2.** Graph the solution to  $x - 2 > 1$  on a number line.

- (A)  $(-\infty, 3)$
- (B)  $(-\infty, -3)$
- (C)  $(-3, \infty)$
- (D)  $(-\infty, 1)$
- (E)  $(3, \infty)$

**Q3.** Write the solution to  $x + 4 \leq 9$  in interval notation.

- (A)  $[-4, 5)$
- (B)  $(-\infty, 5]$
- (C)  $[5, \infty)$
- (D)  $(-\infty, -5)$
- (E)  $[-5, \infty)$

**Q4.** Solve the inequality  $-3x + 2 < 5$ .

- (A)  $x > -1$
- (B)  $x < 1$
- (C)  $x < -1$
- (D)  $x > 1$
- (E)  $x = -1$

**Q5.** Graph the solution to  $-2x - 3 \geq -5$  on a number line.

- (A)  $(-\infty, 1)$
- (B)  $(-\infty, -1]$
- (C)  $[-1, \infty)$
- (D)  $(-1, \infty)$
- (E)  $[1, \infty)$

**Q6.** Solve the inequality  $\frac{2}{3}x - 1 > 0$ .

- (A)  $x > \frac{3}{2}$
- (B)  $x < \frac{3}{2}$
- (C)  $x > -\frac{3}{2}$
- (D)  $x < -\frac{3}{2}$
- (E)  $x = \frac{3}{2}$

**Q7.** Write the solution to  $x - 2 > 4$  in interval notation.

- (A)  $(6, \infty)$
- (B)  $(-\infty, 6)$
- (C)  $(-6, \infty)$
- (D)  $(-\infty, -6)$
- (E)  $[6, \infty)$

**Q8.** Solve the inequality  $-x + 3 > 2$ .

- (A)  $x < 1$
- (B)  $x > 1$
- (C)  $x > -1$
- (D)  $x < -1$
- (E)  $x = 1$

### Open Ended Questions (FRQ)

**Q9.** Solve the inequality  $2x^2 + 5x - 3 > 0$  and graph the solution on a number line. Write your answer in interval notation.

**Q10.** Solve the inequality  $x^2 - 4x - 3 \leq 0$  and write the solution in interval notation. Then, graph the solution on a number line.

### Chef's Tips

- Remind students to check their work by plugging solutions back into the inequality.
- Emphasize the importance of understanding the properties of inequalities.
- Encourage students to use a number line to visualize the solutions.